

LESSON 5-2 • SESSION 2 • PRACTICE • FORM A

Missing Measures in Triangles

Learning target – I can find a missing base or height of a triangle when I know its area. · 6.G.A.1

REMEMBER

- Work backward from $A = \frac{1}{2} \times b \times h$.
- **Double the area** to undo the half, then **divide** by the measure you know.
- Check: $\frac{1}{2} \times b \times h$ gives the area. A base or height is in cm, not cm^2 .

WORKED EXAMPLE

Area 40 cm^2 , base 10 cm . Find the height.

Step 1 $40 = \frac{1}{2} \times 10 \times h$

Step 2 $2 \times 40 = 80$

Step 3 $80 \div 10 = 8$

Answer 8 cm

WORDS area (*área*) · formula (*fórmula*) · inverse operation (*operación inversa*)

PRACTICE

show your work · label units

1. Find the missing measure of each triangle.

a. $A = 18 \text{ cm}^2$, $h = 4 \text{ cm}$; $b = \underline{\hspace{2cm}}$

b. $A = 36 \text{ in}^2$, $b = 9 \text{ in}$; $h = \underline{\hspace{2cm}}$

c. $A = 50 \text{ m}^2$, $h = 10 \text{ m}$; $b = \underline{\hspace{2cm}}$

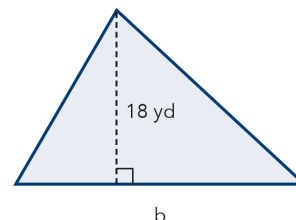
d. $A = 63 \text{ ft}^2$, $b = 14 \text{ ft}$; $h = \underline{\hspace{2cm}}$

2. **Park Lot.** A triangular lot covers 126 yd^2 . Its height is 18 yd .

Hint Double the area first.

a. Fill in the formula: $126 = \frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

b. Find the base b . $\underline{\hspace{2cm}} \text{ yd}$



3. **Quilt Piece.** A triangular quilt piece has an area of 54 in^2 and a base of 12 in . Find the height.

Answer $\underline{\hspace{2cm}} \text{ in}$.

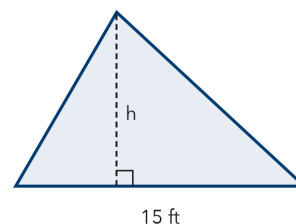
4. **Roof Gable.** A triangular gable covers 117 ft^2 . Its height is 9 ft . Find the base.

Answer $\underline{\hspace{2cm}} \text{ ft}$

5. **Awning Panel.** The awning panel covers 60 ft^2 . Find the height h .

Hint Undo the $\frac{1}{2}$ by doubling 60 .

Answer $\underline{\hspace{2cm}} \text{ ft}$



- 6. Sign Claim.** A triangular sign has an area of 84 cm^2 and a height of 7 cm. Rosa says the base must be 28 cm. Is she right? Use the formula.

Hint Put Rosa's base into $\frac{1}{2} \times b \times h$ and check.

Show your work.

Base _____

- 7. Sandbox Wedge.** A triangular wedge of a sandbox covers 45 m^2 . Its height is 5 m. Find the base.

Show your work.

Answer _____ m

- 8. Two Signs.** Sign A is a triangle with area 120 ft^2 and base 16 ft. Sign B is a triangle with area 120 ft^2 and height 12 ft.
- Find the height of Sign A. _____ ft
 - Find the base of Sign B. _____ ft
 - Both signs have the same area. How can their shapes be different?

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Bennett's Work.** A triangle has an area of 66 ft^2 and a base of 11 ft. Bennett wrote: " $66 \div 11 = 6$, so the height is 6 ft." Explain his mistake. Then find the height.

TEST PRACTICE

choose the best answer

- 10.** A triangle has area 60 cm^2 and base 10 cm. Its height is –
- A** 6 cm **B** 12 cm **C** 50 cm **D** 600 cm
- 11.** A triangle has area 45 in^2 and height 9 in. Its base is –
- A** 5 in. **B** 36 in. **C** 10 in. **D** 405 in.

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

13 Z	14 M	26 S	20 G	7 Q	9 I	24 I	18 N	4 X	8 S
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#2b	#3	#4	#5	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____